Nandam Josh Nihanth

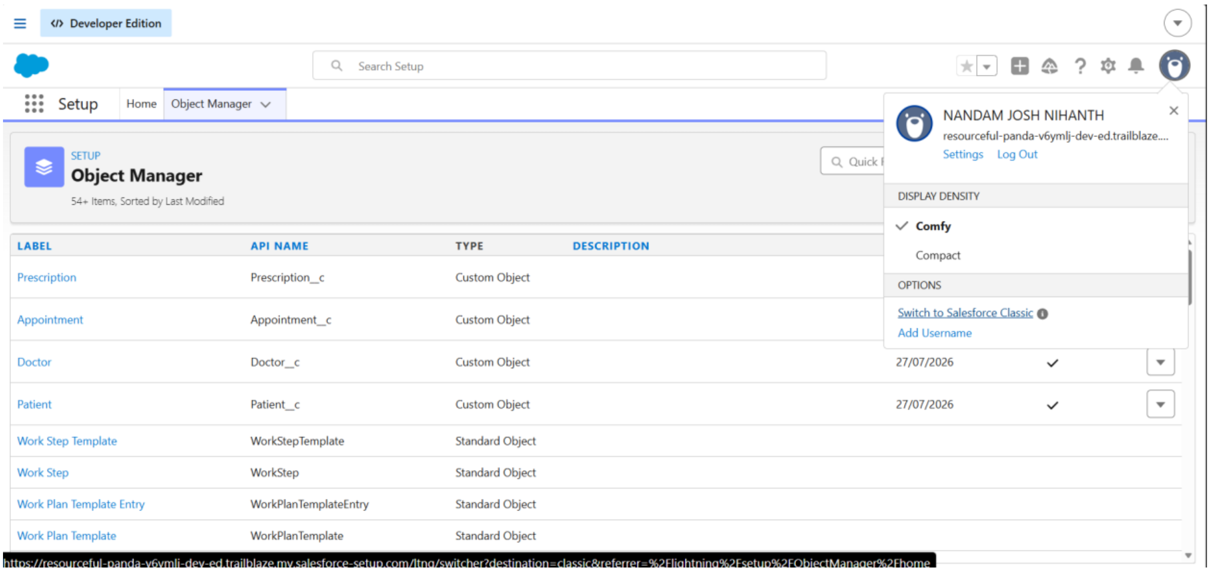
23PA1A05H1

CSE-C

Salesforce Day 1 tasks

Data Modeling And Object Relationships :

Task 1 — Design a mini data model



Bacically object are 2 types

1.Standard Objects

2.Custom objects

1.Standard Objects :

That are created default by the salesforce like Accounts, lead, contacts, oppournuintys,price Book and soon these are comes under the Standard Objects.

2.Custom objects :

The Custom objects that are created by ourselfs and it should be like Name_C , age_c and soon these are the Custom objects.

In this I have created the 4 custom object

1.Doctor

2.Patient

3. Appointment

4.presceptions

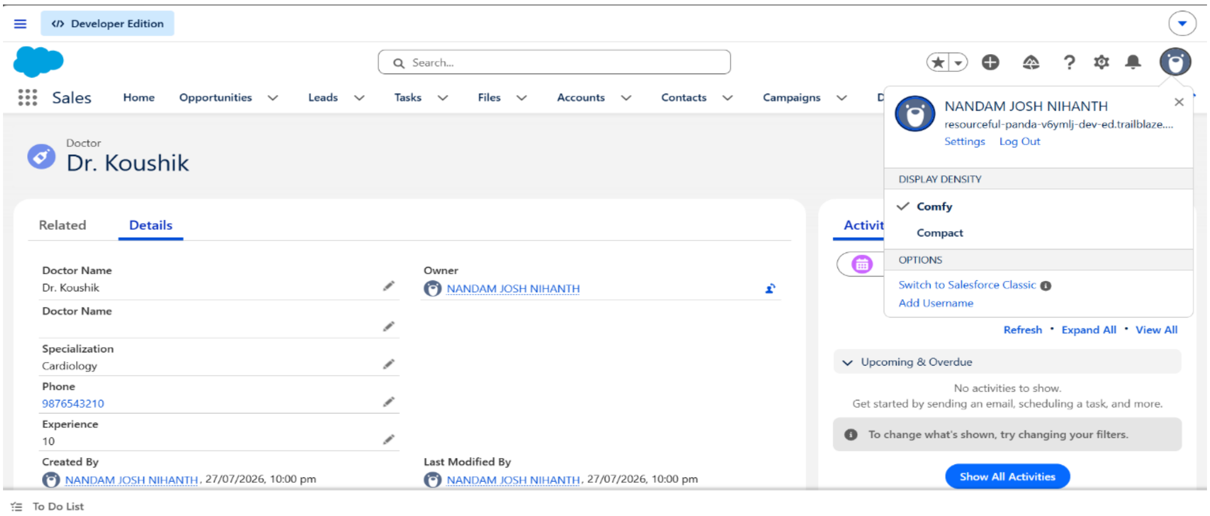
In the each Custom objects we have created the different fields
For the Doctor We have created the

1. Doctor Name (Text)

2. Specialization (Text)

3. Phone (Phone)

4. Experience (Number)



For the Patients We have created the

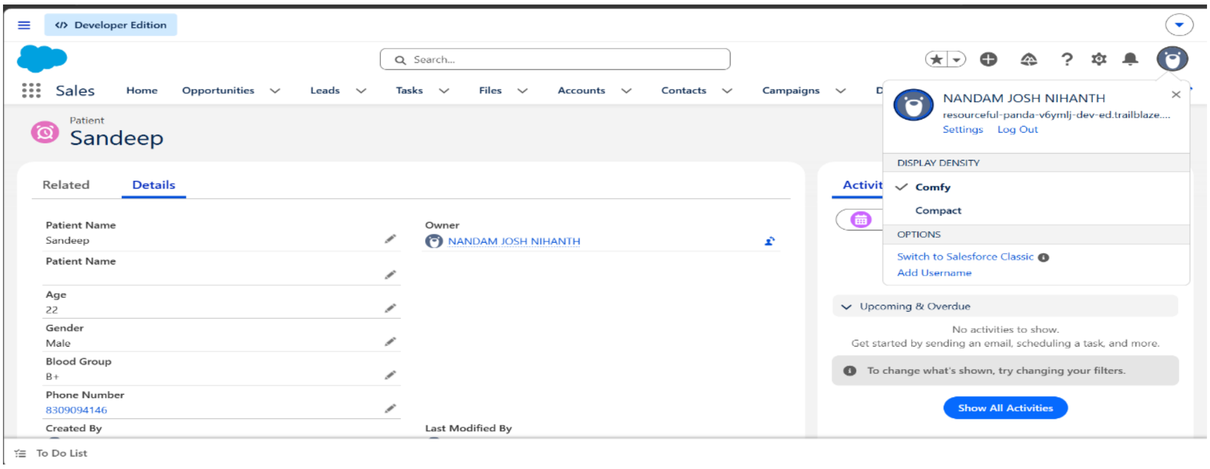
1. Patient Name (Text)

2. Age (Number)

3. Gender (Picklist)

4. Phone (Phone)

5. Blood Group (Picklist)

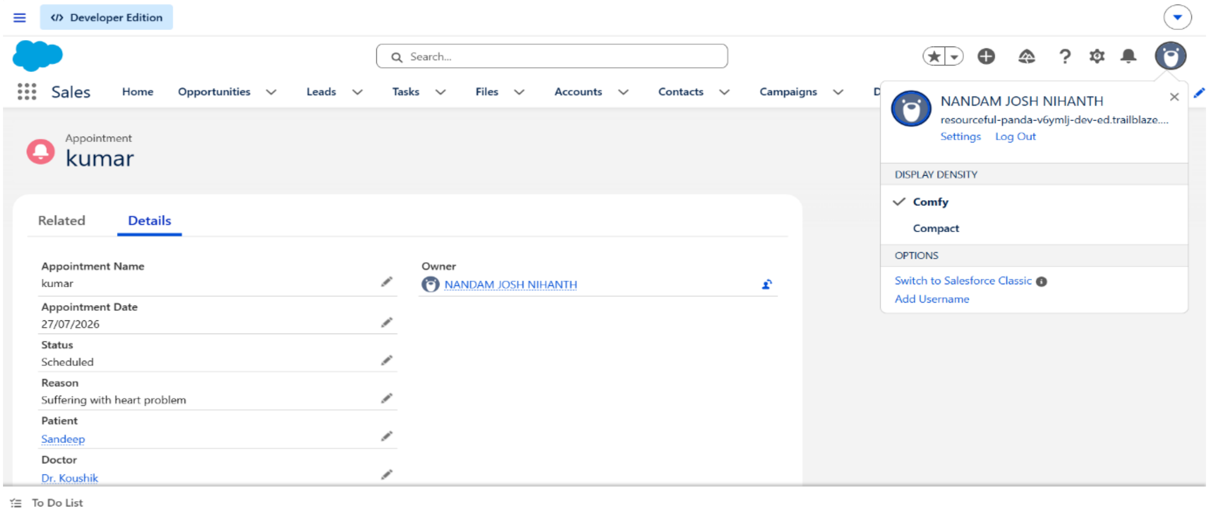


For the Appointment We have created the

1. Appointment Date (Date)

2. Status (Picklist: Scheduled, Completed, Cancelled)

3. Reason (Text Area)

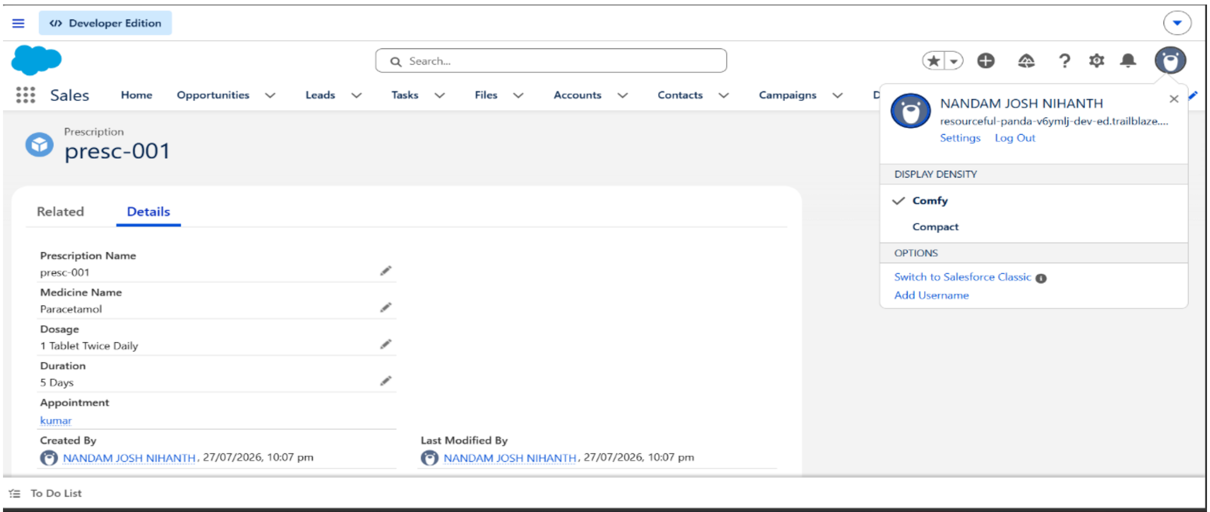


For the Prescription we have created the

1.Medicine Name (Text)

2.Dosage (Text)

3.Duration (Text)



After that created some randam data for the each field

For example:

Doctor Name :Dr .Koushik   Phone :9876543210

Specialization : Cardiology   Experience :10

Patient Name:Sandeep

Age: 18

Gender: Male

Phone :963824710

Blood Group: B+

Appointment: Rahul's Appointment

Medicine: Paracetamol

Dosage: 1 Tablet Twice Daily

Appointment Date :28/0/2026

Status :completed

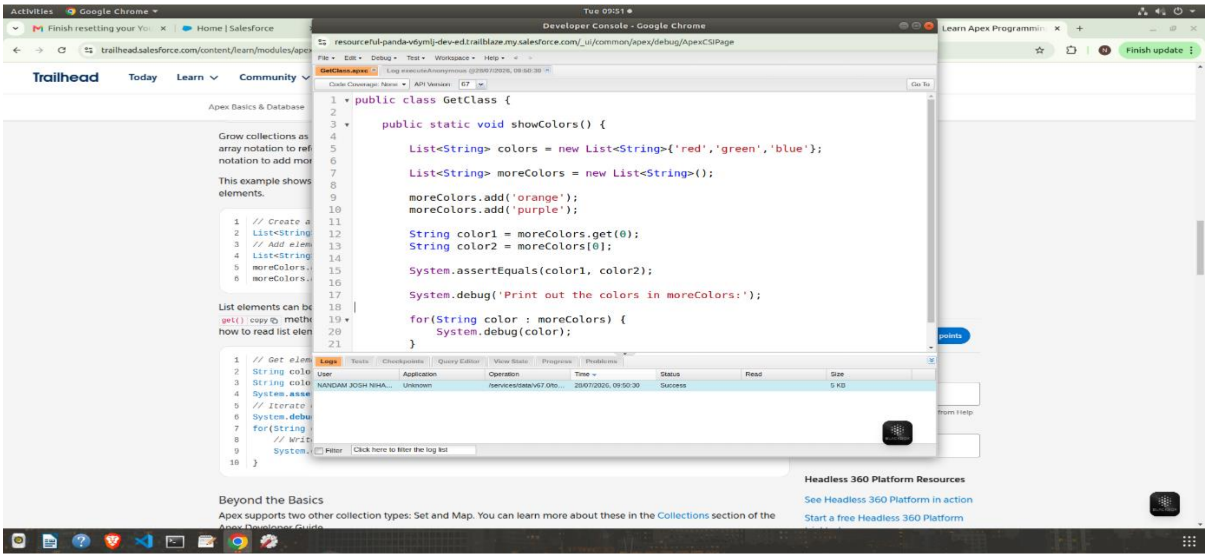
Reason :Fever

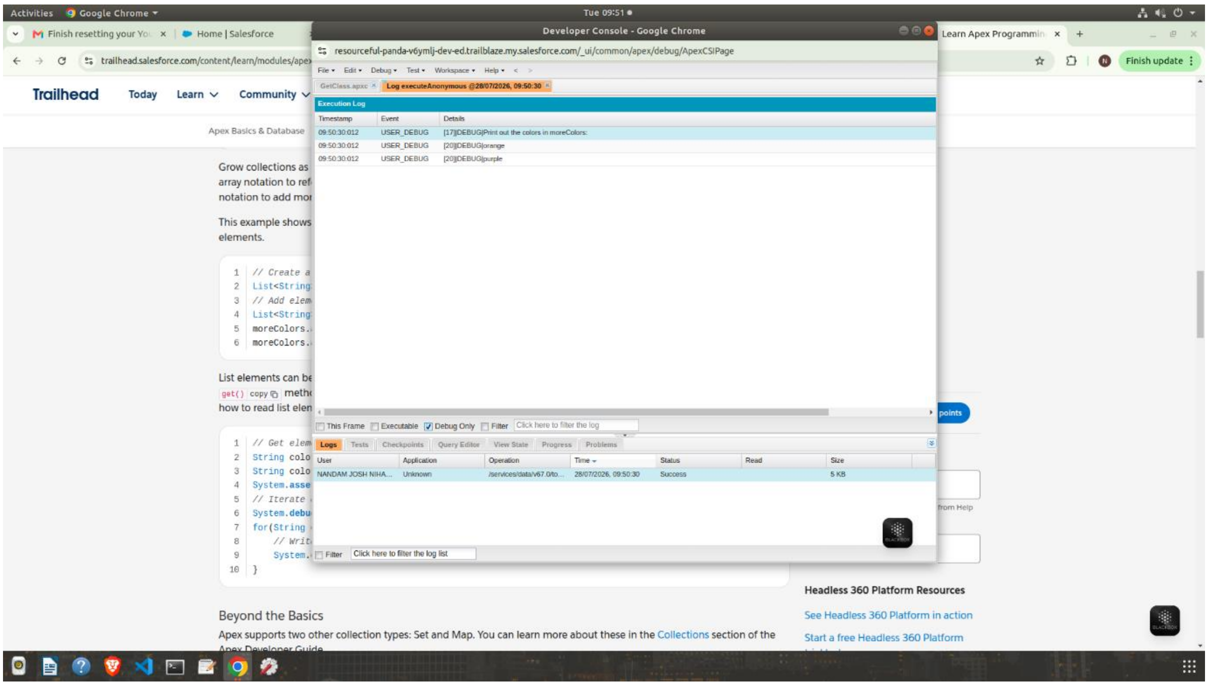
Successfully designed and implemented the Hospital OPD data model with custom objects, relationships, and sample records.

Task 2 — Write it yourself

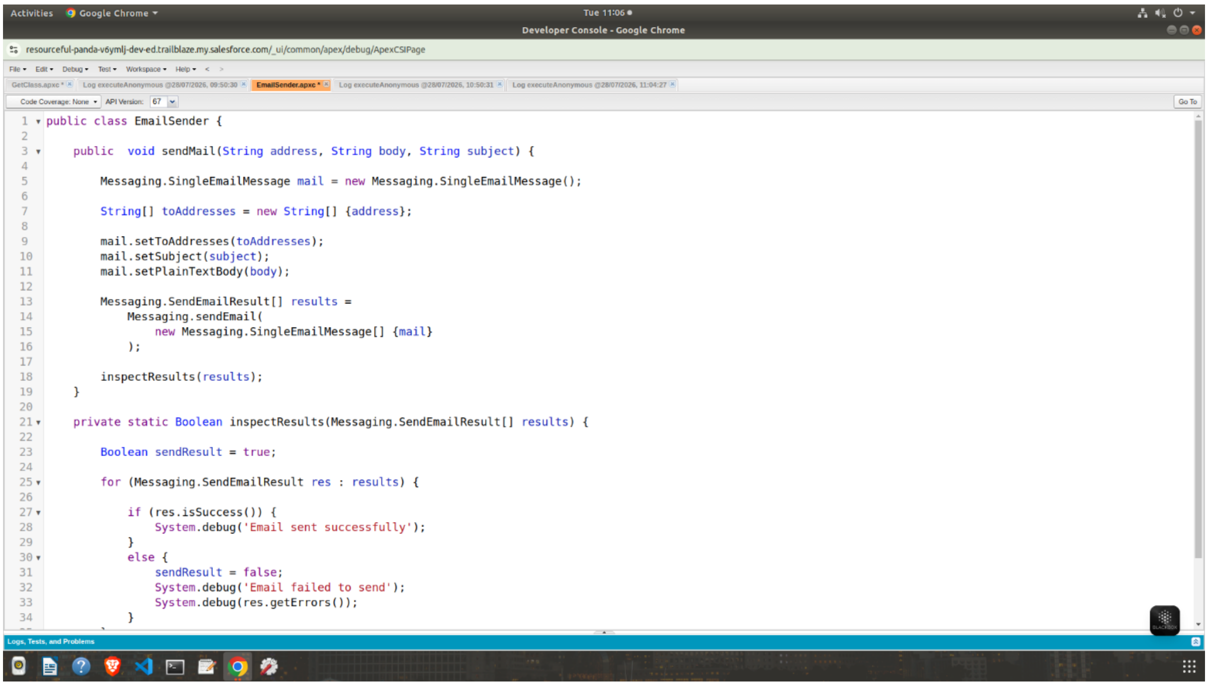
Type out every code example from the module by hand in Execute Anonymous (Debug menu, Developer Console), run it, read the debug log. Don't just click the Trailhead checker without running it yourself first

For this we have created the apex class the below screenshot show the basic program of the apex language for the salesforce we use the apex language this module shows about the entire about the developer bases





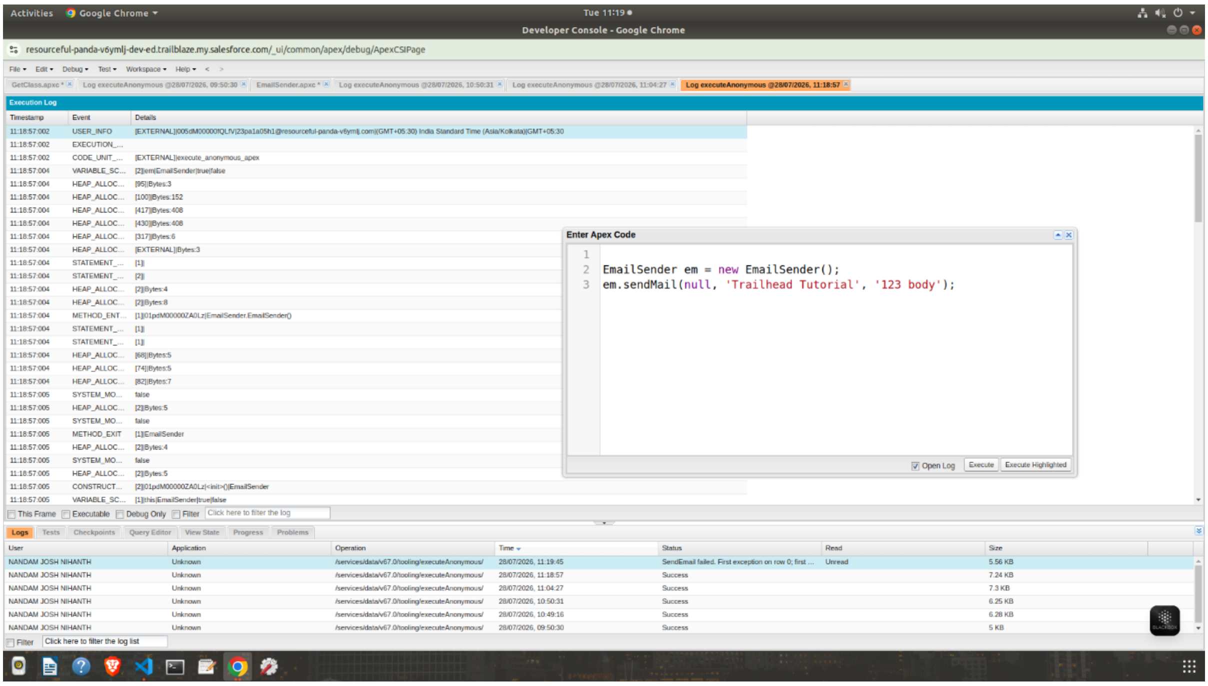
This shows the output of the morecolors program



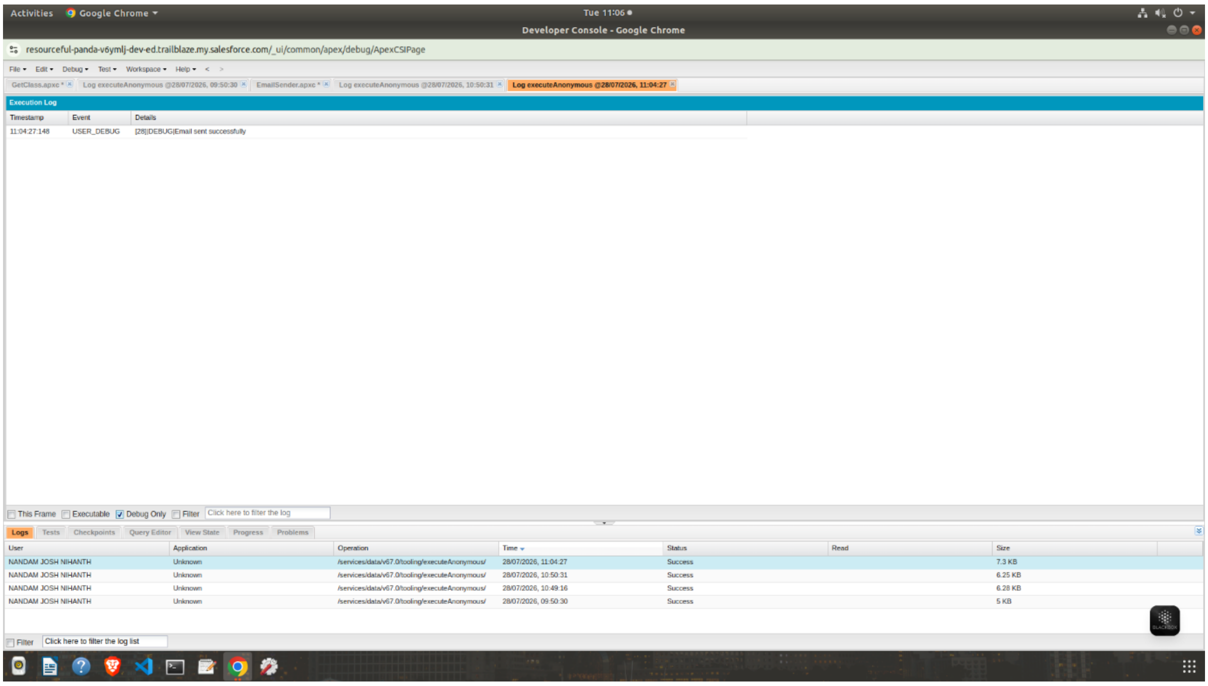
sss

While entiring the data like mail id and body subject that the execution is not possible for the null values

And the we have to entire the perfect email id for the verification



After providing the email correctly the output has shown correctly

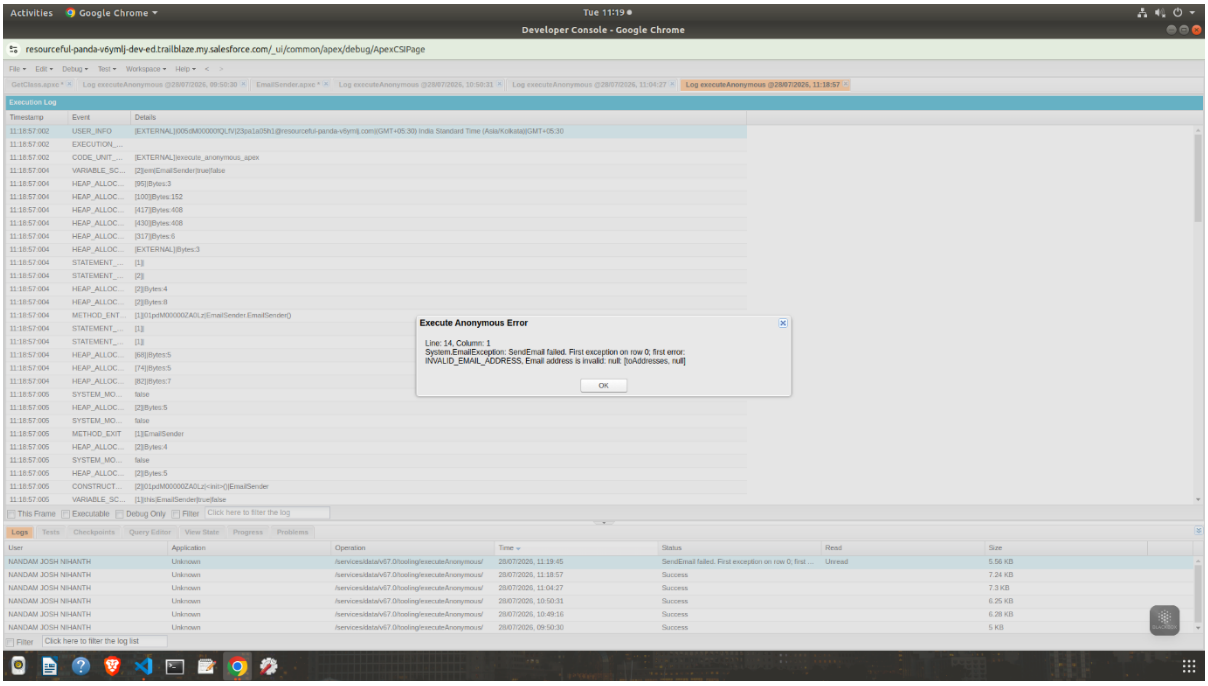


Therefore the email has been sent successfully

And the given mail id joshnihanthnandam@gmail.com

The below image shows the error that was occur while entering the null values and text values

We need to give the perfect data while executing the program



Successfully executed basic Apex programs using the Execute Anonymous window and verified the results through the Debug Log.

Task 3 — SOQL practice

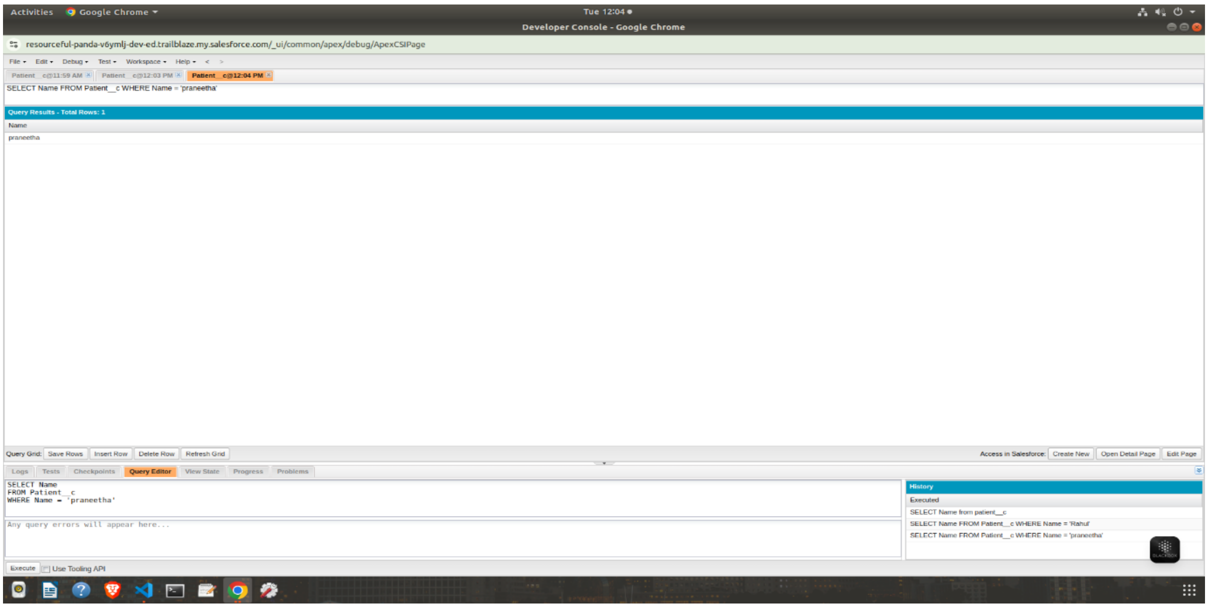
The SOQL stands for the salesforce object query language and it have the SELECT, WHERE, ORDER BY, LIMIT, relationship queries, and aggregate functions

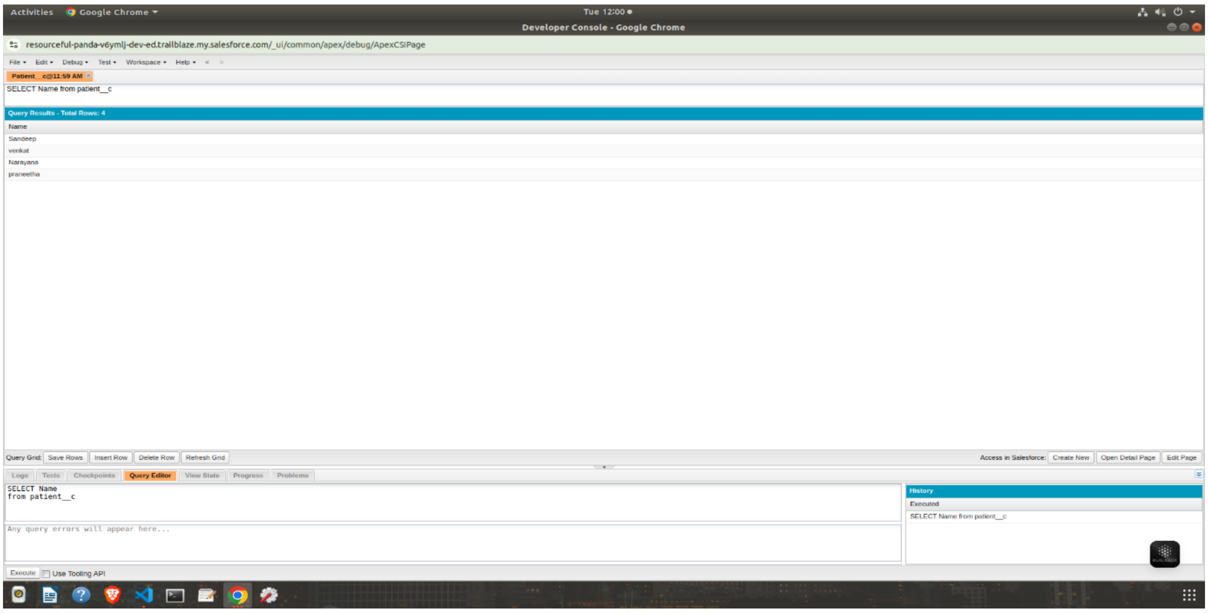
The SOQL was Executed in the Developer console and in the query Editor while executing it shows the output of the each query.

1.WHERE clause:

The WHERE clause shows the specific condtion in the query.

The WHERE clause helps retrieve only the required records instead of all records.

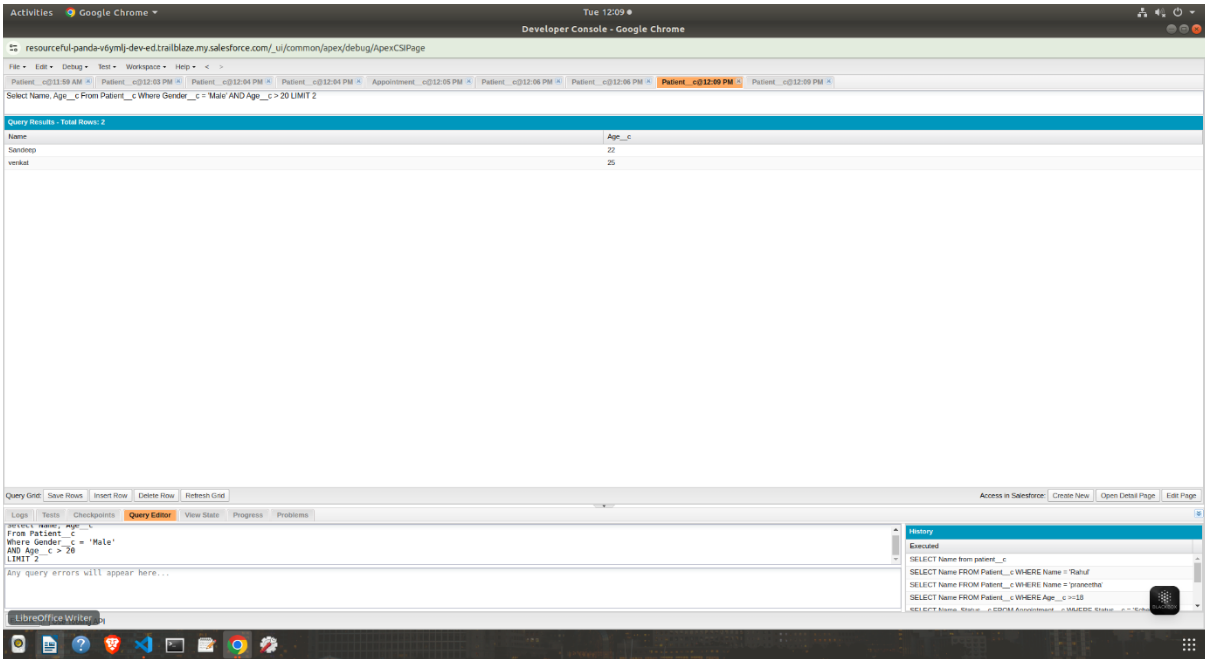




2. ORDER BY + LIMIT

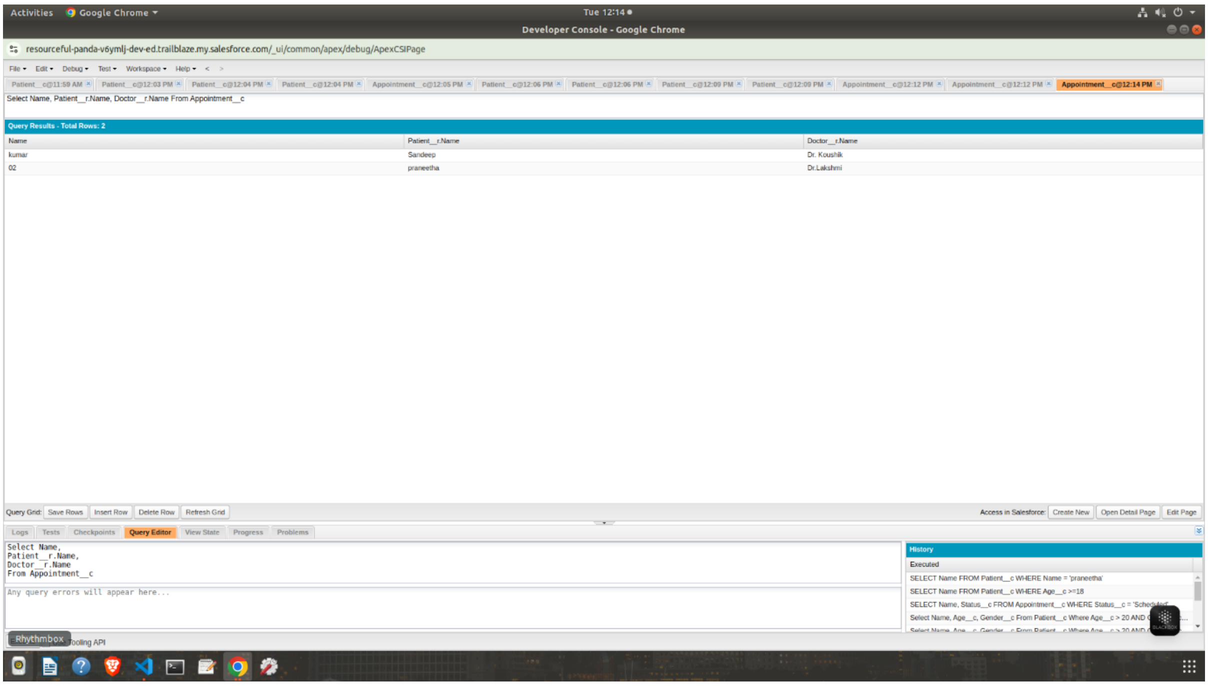
This query returns only the patients whose gender is **Male**.

For this based on the limit it shows the output.



3. Relationship query (parent-to-child or child-to-parent):

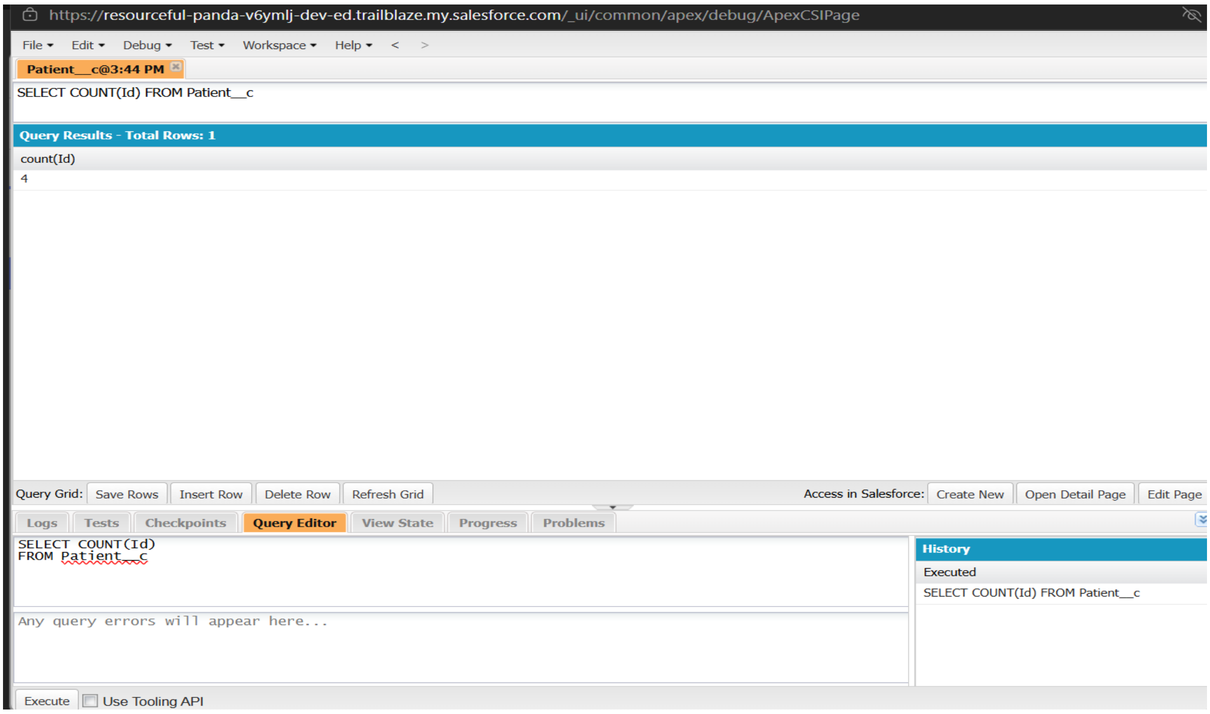
This query retrieves appointment details along with the related patient and doctor names.



4. Aggregate query with COUNT() or GROUP BY

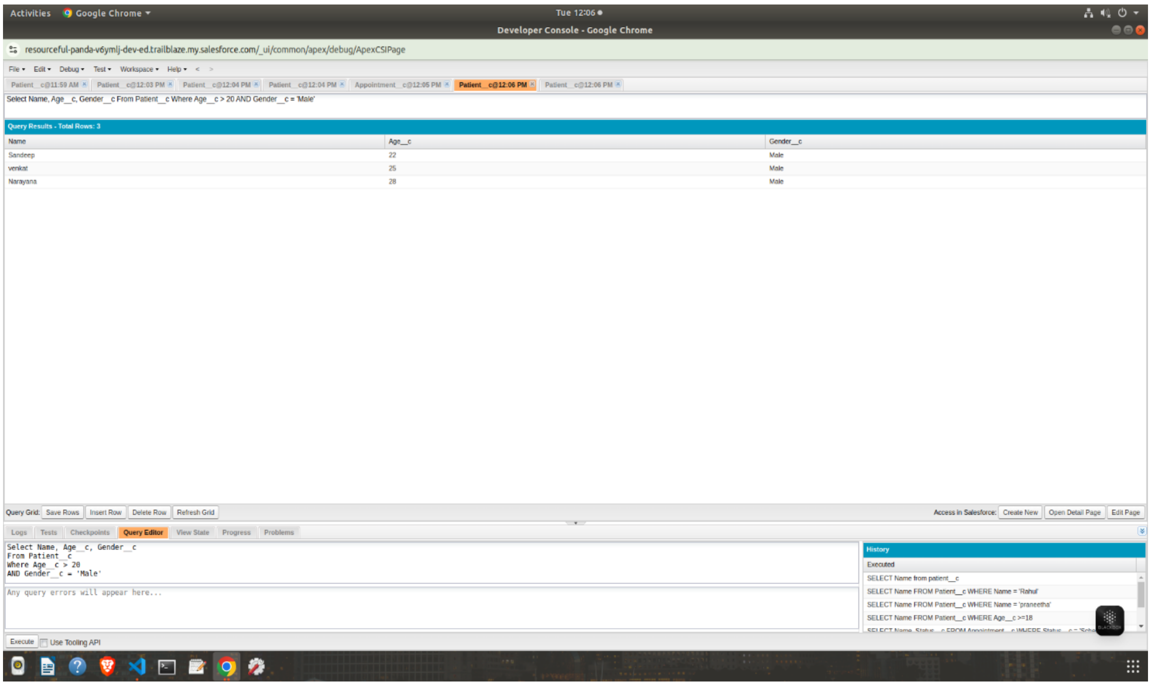
This query uses the COUNT() function to count the number of patient records.

Aggregate functions summarize data instead of displaying individual records.



5. Comparison operator (>=) on a date or number field

Comparison operators such as >=, <=, >, and < are used to filter records based on values.

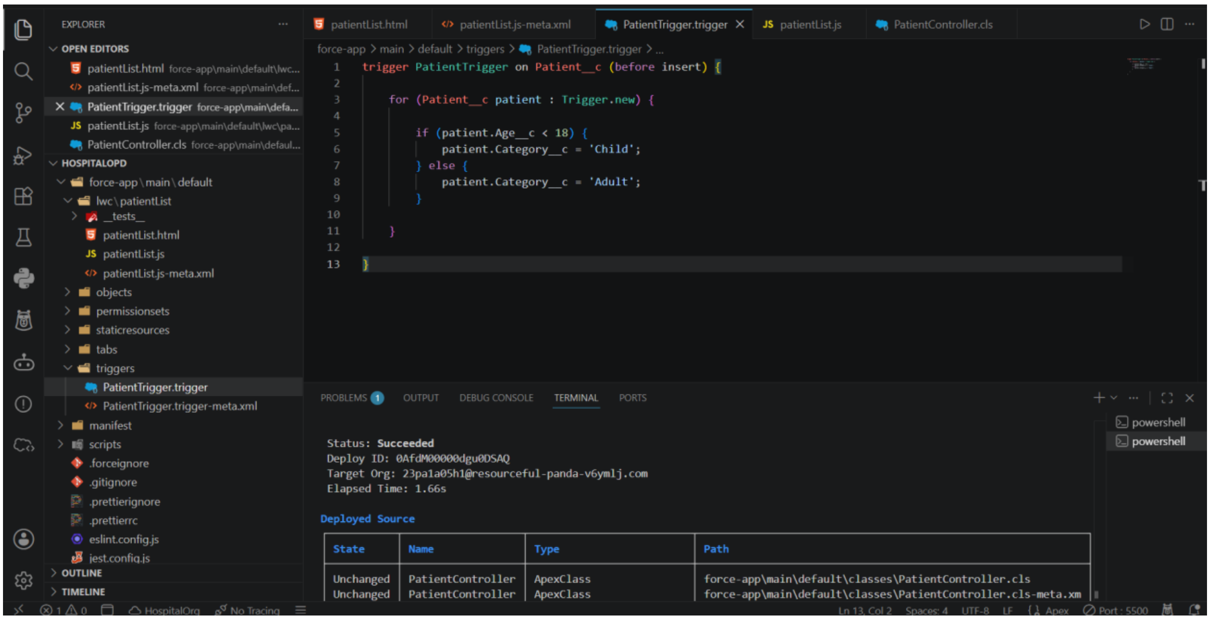


Successfully executed various SOQL queries to retrieve, filter, sort, and count records from the Hospital OPD custom objects.

Task 4: Apex Triggers

An **Apex Trigger** is code that runs automatically when a record is changed in Salesforce.

For apex triggers let take hospital the recepist has store the information of a patient that can be loaded automatically into the fields and shows the entire information about the each patient.



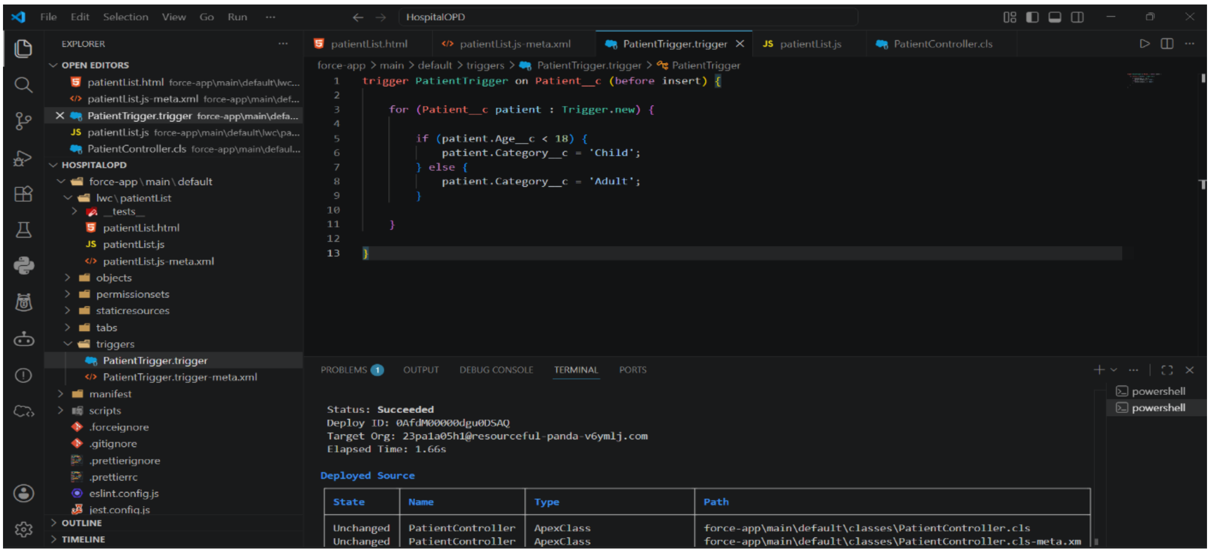
In the patient object I have creaed the field called category that imformation like adult or child

Then I created the apex trigger called PatientTrigger

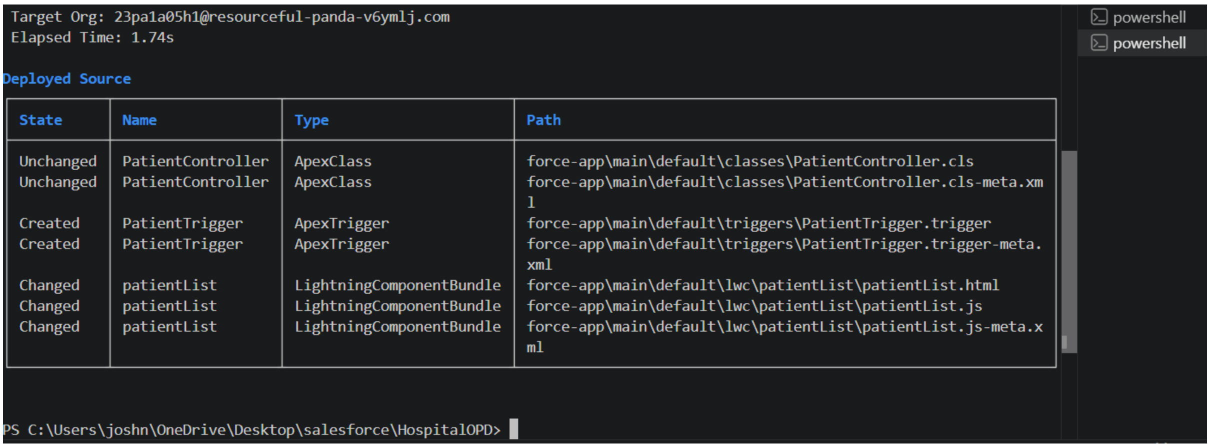
That shows the if age is greater than 18 it shows as the adult

And if is less than 18 then it shows the child

That the overall apex trigger program shows about



I have done the deployment for the patientlist of hospitalopd



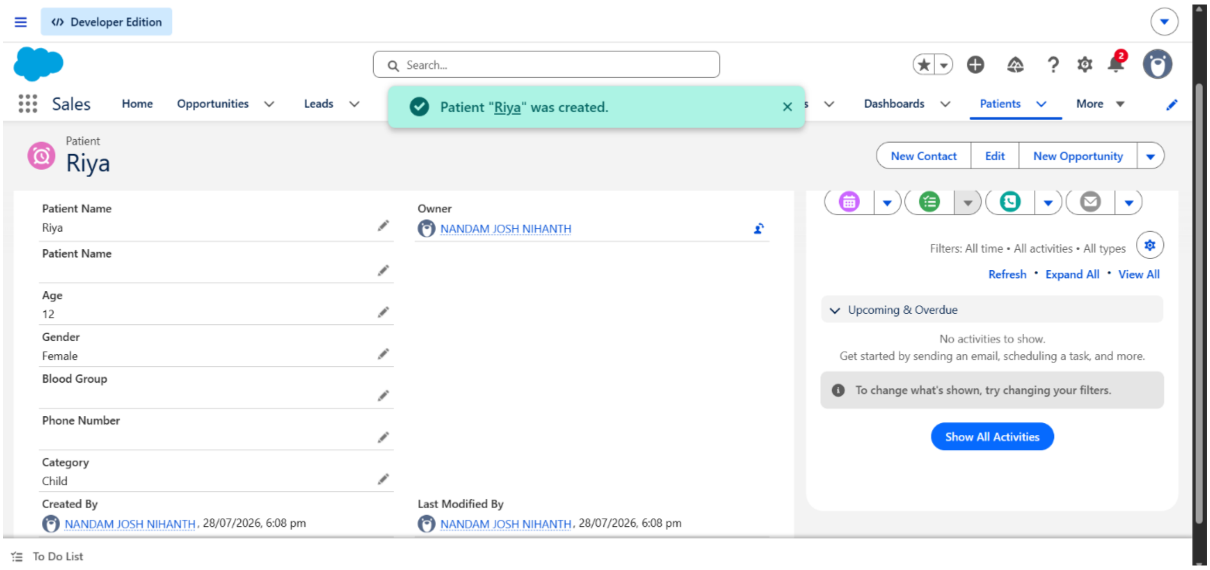
```
Target Org: 23pa1a05h1@resourceful-panda-v6ymlj.com
Elapsed Time: 1.74s

Deployed Source
```

State	Name	Type	Path
Unchanged	PatientController	ApexClass	force-app\main\default\classes\PatientController.cls
Unchanged	PatientController	ApexClass	force-app\main\default\classes\PatientController.cls-meta.xml
Created	PatientTrigger	ApexTrigger	force-app\main\default\triggers\PatientTrigger.trigger
Created	PatientTrigger	ApexTrigger	force-app\main\default\triggers\PatientTrigger.trigger-meta.xml
Changed	patientList	LightningComponentBundle	force-app\main\default\lwc\patientList\patientList.html
Changed	patientList	LightningComponentBundle	force-app\main\default\lwc\patientList\patientList.js
Changed	patientList	LightningComponentBundle	force-app\main\default\lwc\patientList\patientList.js-meta.xml

```
PS C:\Users\joshn\OneDrive\Desktop\salesforce\HospitalOPD>
```

The image clearly shows the deployment of the lighting component



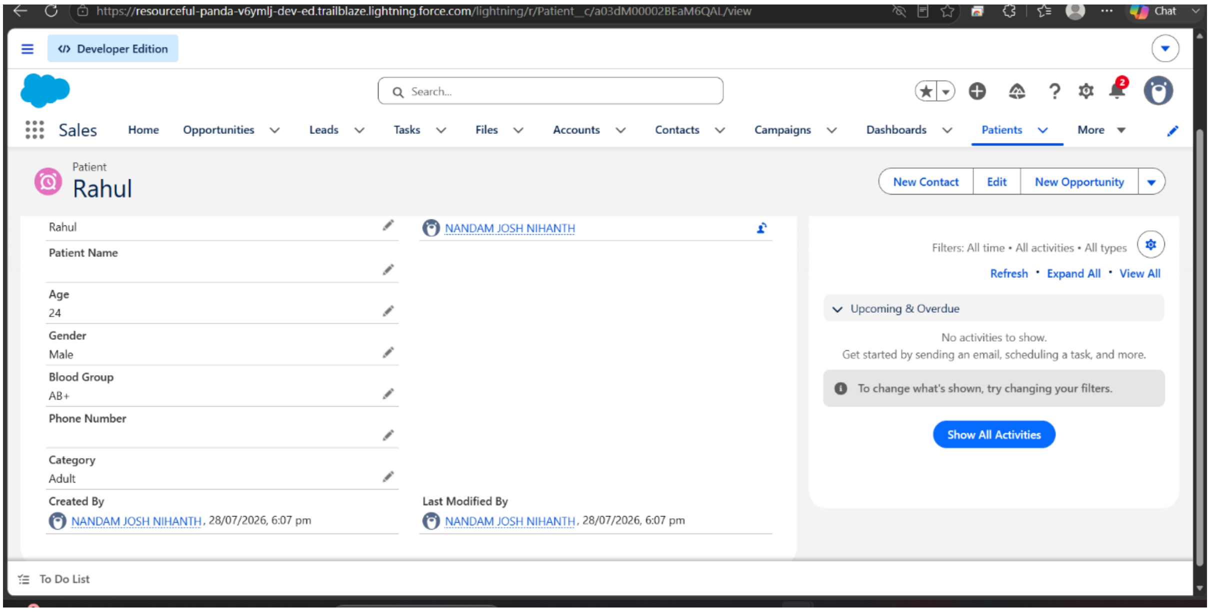
The riya is 12 year old so the category will automatically filled to the adult and the riya is the new patient name that shows the basic information about the patient details of it

In the below image shows the Rahul details

That shows the age that is 18 years old and automatically filled to the adult for the rahuland remainding data has been formal data of the new patient Rahul.

No other fields has been created into that

And other details like bloodgroup, name, age, category and other information of the patients.



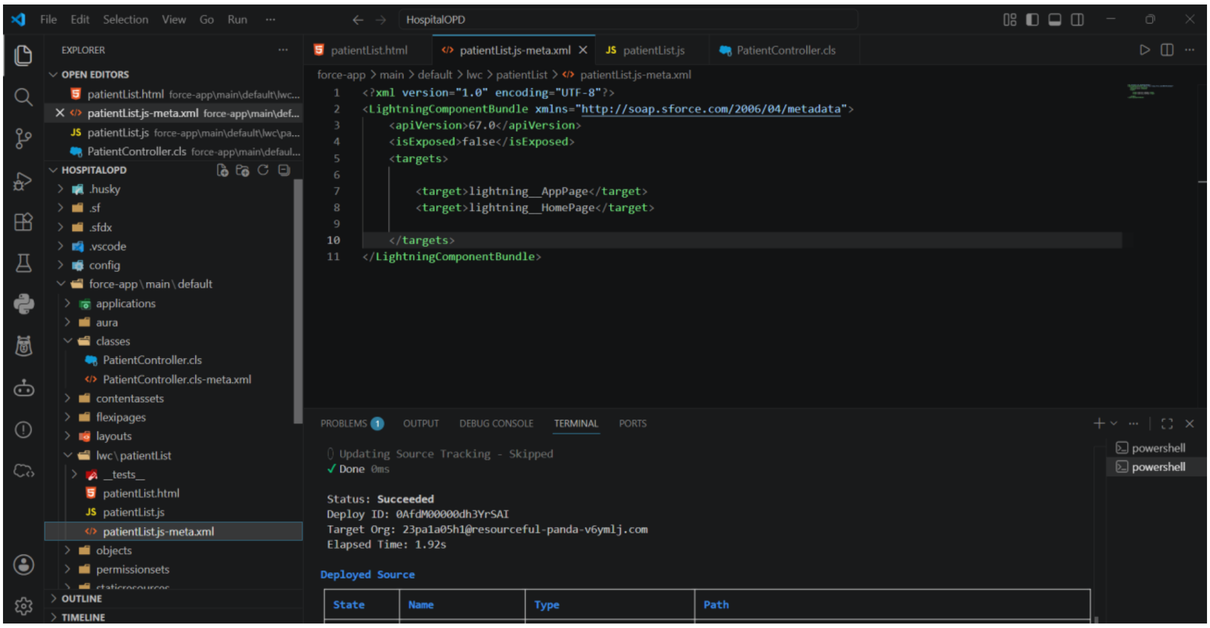
The Apex Trigger was successfully created and deployed to automate the classification of patient records based on age. This task demonstrated how Salesforce triggers can execute business logic automatically before records are saved.

Task 5 — Build one simple component

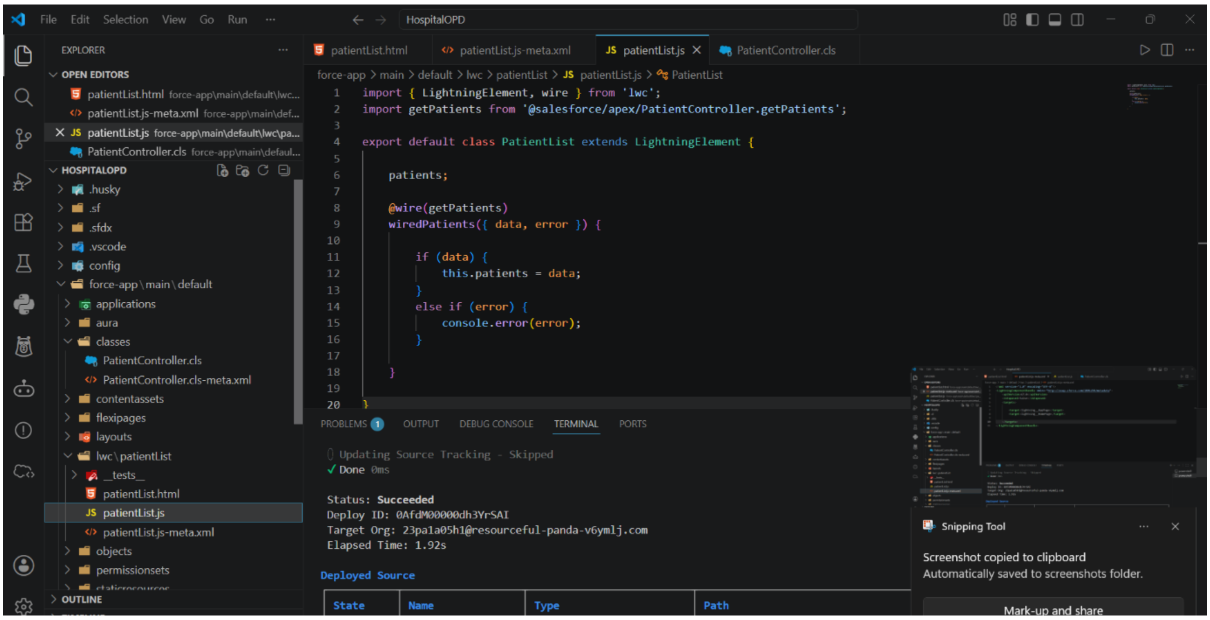
We have to create the lighting web componets for the patients list
By clicking the SHIFT+ CTRL + p we can see the SFDX: Create a lighting web compont

For the name we can set as the paitentlist

Then we have to authorize the org for the connecting with the salesforce
then it will created the html, js, xml file has been created . by linking with the componets



In the xml file that target should shown in the both the app page and home page through the target of the page

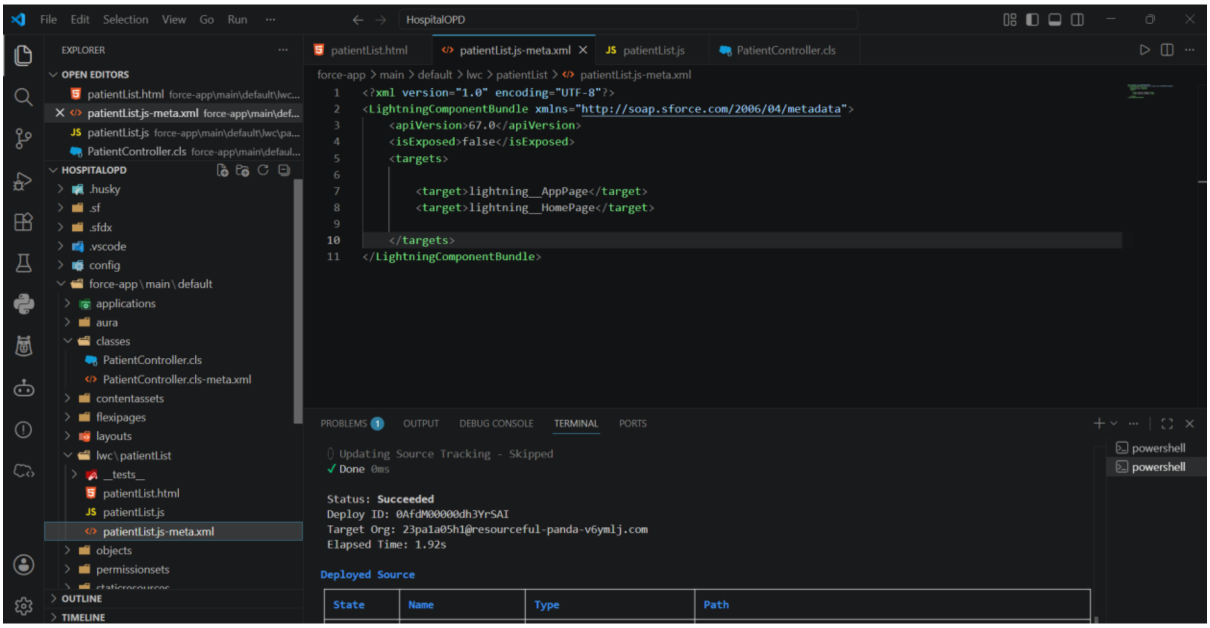


The component consists of three files:

patientList.html

patientList.js

patientList.js-meta.xml



After that the deployment has been started

Command for the deploy : sf project deploy start

By clicking enter the deployment has been started

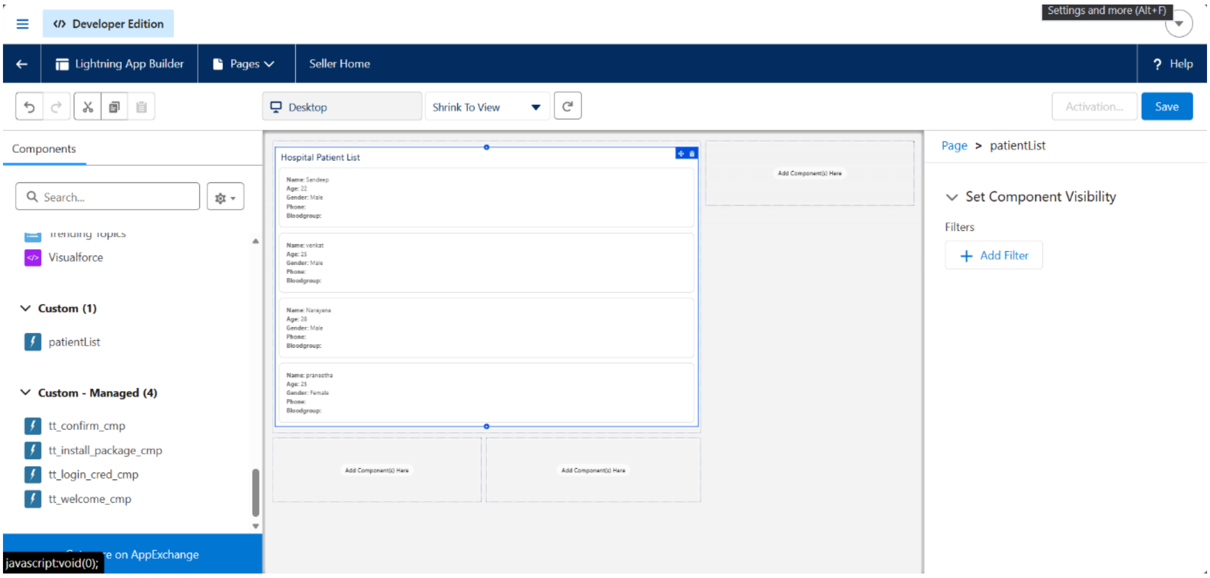
```
Updating Source Tracking - Skipped
✓ Done 0ms

Status: Succeeded
Deploy ID: 0AfdM00000dh3YrSAI
Target Org: 23pa1a05h1@resourceful-panda-v6ymlj.com
Elapsed Time: 1.92s

Deployed Source
```

State	Name	Type	Path
Unchanged	PatientController	ApexClass	force-app\main\default\classes\PatientController.cls
Unchanged	PatientController	ApexClass	force-app\main\default\classes\PatientController.cls-meta.xml
Created	patientList	LightningComponentBundle	force-app\main\default\lwc\patientList\patientList.html
Created	patientList	LightningComponentBundle	force-app\main\default\lwc\patientList\patientList.js
Created	patientList	LightningComponentBundle	force-app\main\default\lwc\patientList\patientList.js-meta.xml

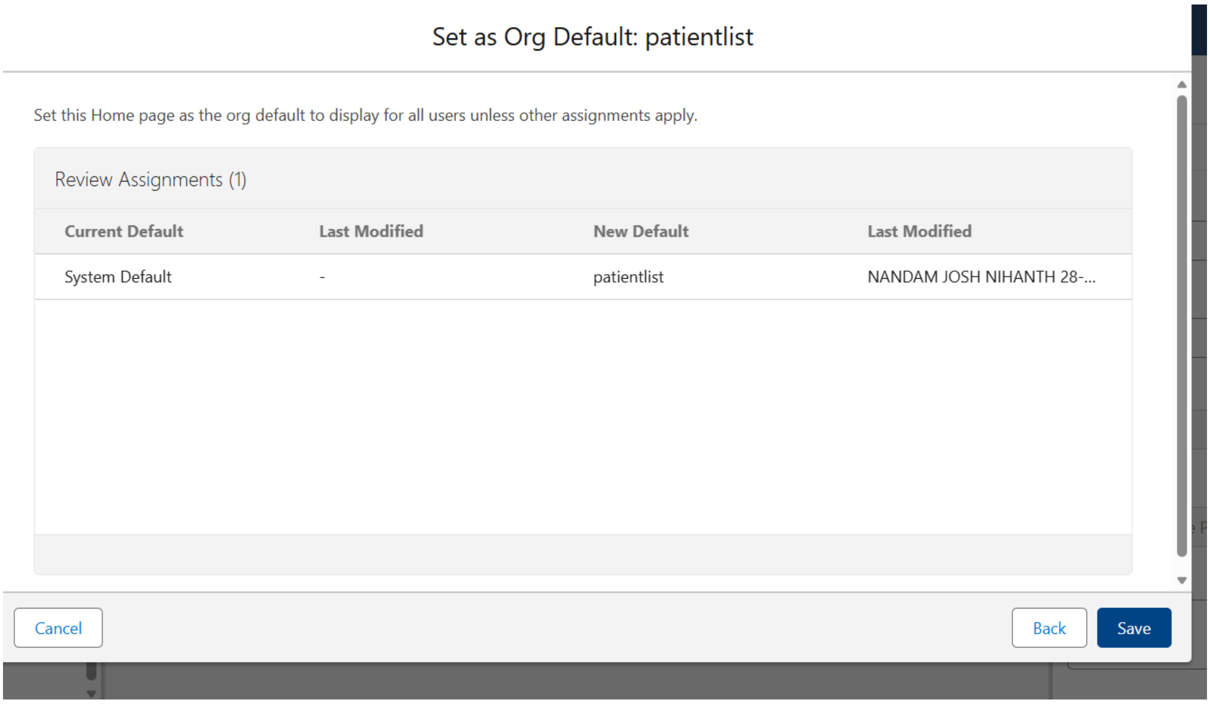
After that the custom component has shown in the edit page we have to drag the patientlist into the main page as the component and the overall data has been appear.



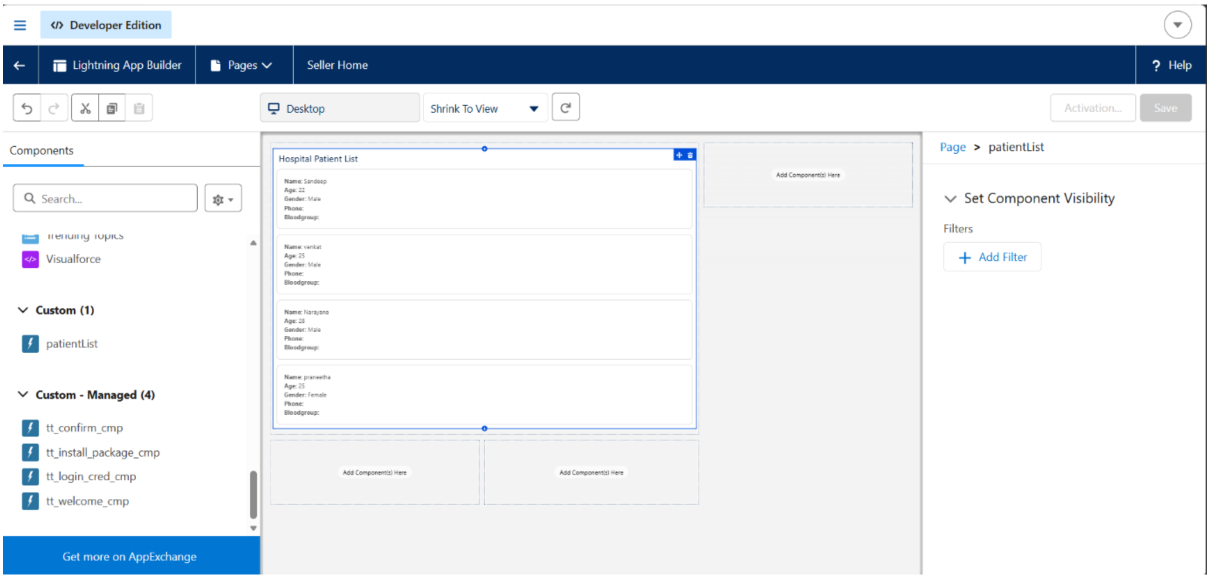
By clicking save

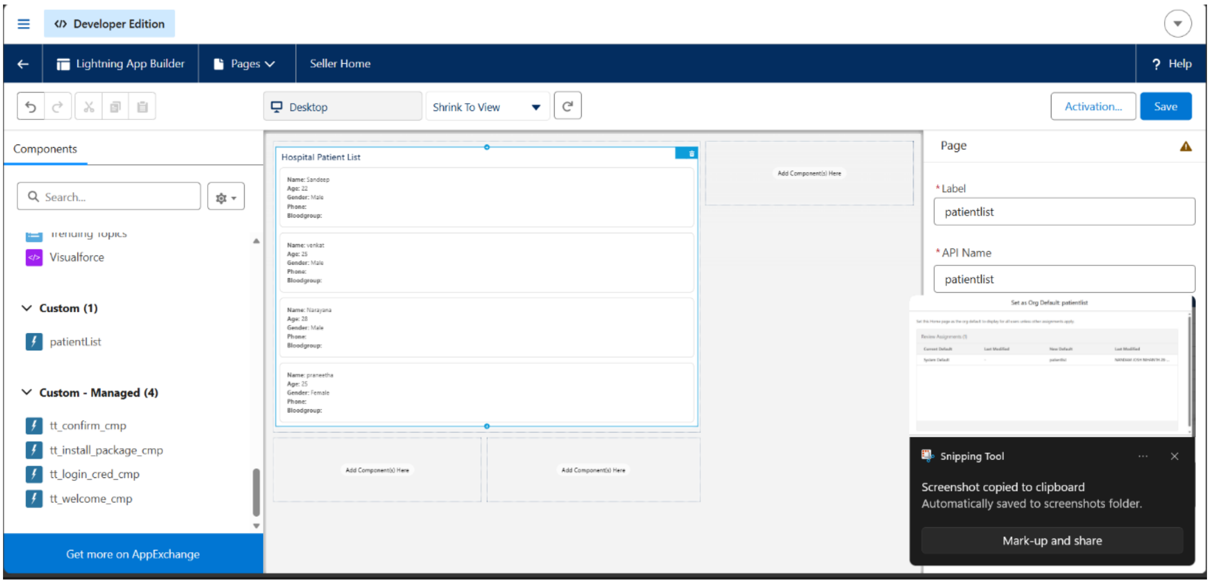
And then we need to activate the page by clicking the Activate button

Then we need to assign to the org and set as the default for the other file



Then click save for the





The Lightning Web Component was successfully developed, deployed, and added to the Salesforce Home Page to display patient records dynamically. This task improved my understanding of Lightning Web Components, Apex integration, and data retrieval using the @wire service.
